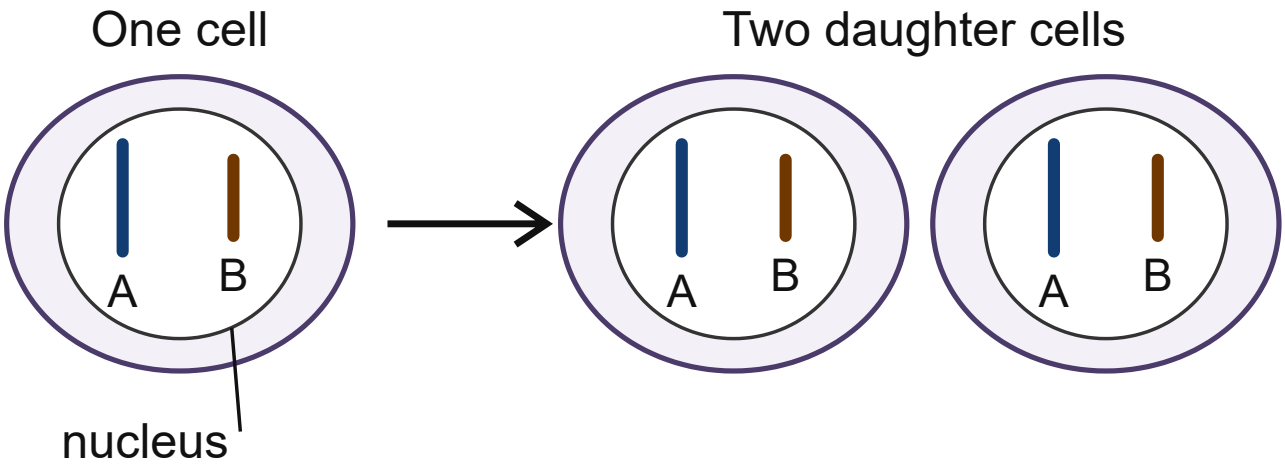
# Reference A matching chromosome sets




Compare the full A and B set in both daughter cells. Each new cell has a matching set.

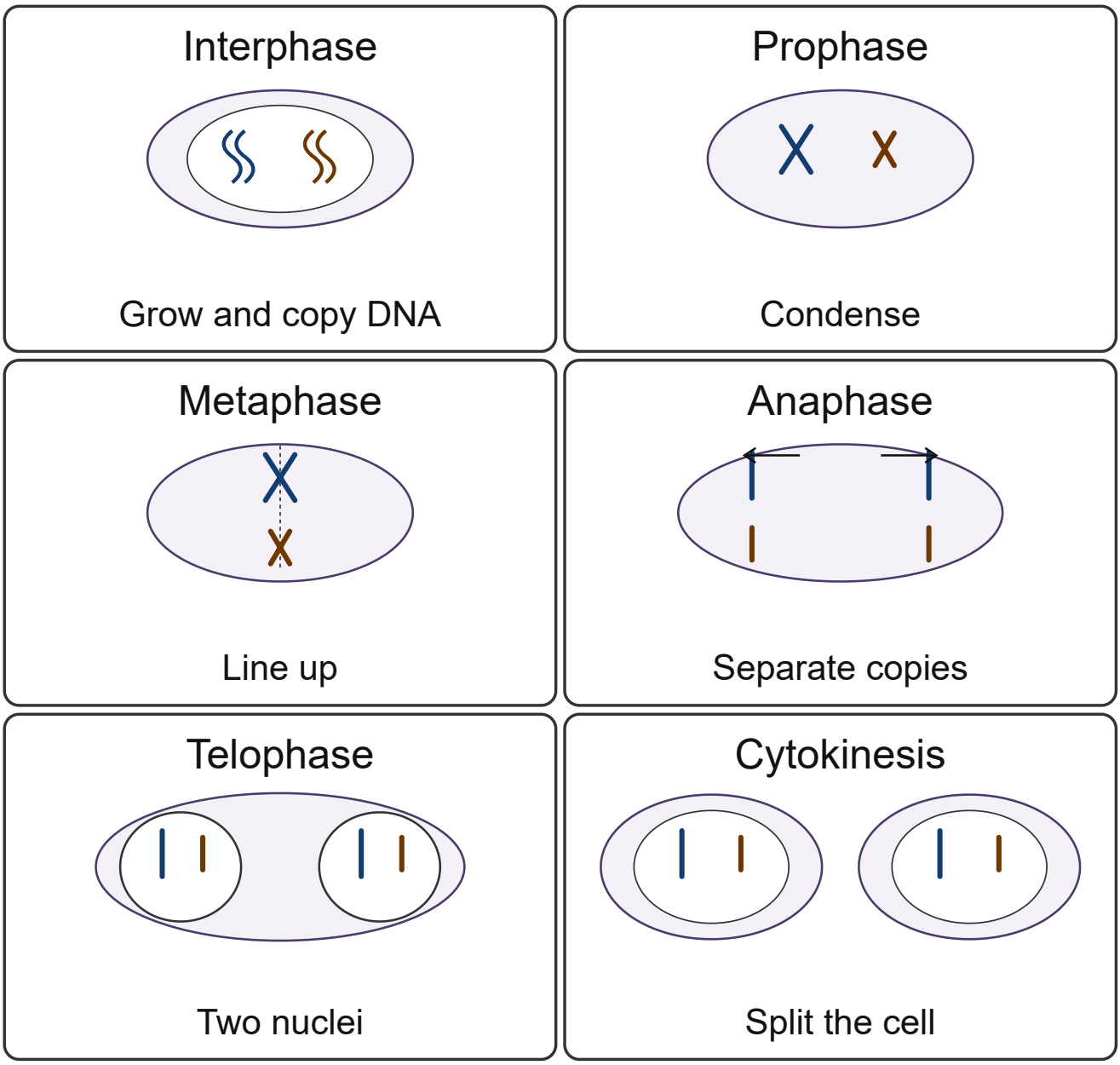
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

| Step | What changes |
|---|---|
| Interphase | The cell grows and copies its DNA. |
| Mitosis | Copied chromosomes separate into two nuclei. |
| Cytokinesis | The cell splits into two daughter cells. |

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

Made by Matt

# Reference B the six named stages




Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

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# Arrival and exit Aspire

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Using the picture alone, state the number of separate cells after the final split.

2. Which labelled cell structure contains the chromosome set?

3. Explain what the A and B labels show about the sets in the two new cells.

4. Give one change shown by the model and one thing its simplified symbols do not show.

## Exit ticket

1. A model skips copying and later shows uneven sets. Which error should you repair first, and why?

2. Why would a correct final cell split fail to fix uneven chromosome separation?

**Pupil voice**

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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# You do Aspire

Use your book or this sheet. Use Reference A and B. One response per task.

**1. Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]**
**Staff I ☐ S ☐ C ☐**

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**2. Explain why swapping mitosis and DNA copying would break the model. [3]**
**Staff I ☐ S ☐ C ☐**

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**3. Predict what would happen to the sets if copied chromosomes did not separate evenly. Give a reason. [3]**
**Staff I ☐ S ☐ C ☐**

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# Feedback and response · Aspire

## Feedback codes

| Code | Meaning |
|---|---|
| VF | Verbal feedback |
| WS | Worked with support |
| I | Independent |
| NS + … | Next step |
| E | Evidence captured |
| R | Pupil responded to feedback |
| // | Self-edit point |
| ? | Read this back to me |

## My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt